

Epistemic Blind Spots in World-Model-Based Active Inference: Why RSSM Imagination Cannot See Obstacles

Jaerock Kwon

Department of Electrical and Computer Engineering
University of Michigan–Dearborn
jrkwon@umich.edu

Abstract

We investigate whether obstacle avoidance can emerge purely from Expected Free Energy (EFE) minimization in a deep active inference agent for autonomous driving—without privileged obstacle coordinates. Using an RSSM world model trained on CARLA highway scenarios, we systematically test whether learned preferences can drive obstacle avoidance through imagined trajectory planning. We discover a fundamental **epistemic blind spot**: while the RSSM posterior (filtering) correctly encodes obstacle presence, the RSSM prior (imagination) remains obstacle-blind. A contrastive preference model achieves 37 nats separation between obstacle-free and obstacle-present posterior latents; an auxiliary obstacle prediction head achieves *perfect* binary separation (0.000 vs 1.000) and amplifies contrastive separation to 276 nats. Yet obstacle avoidance remains at 0% in all cases—the RSSM transition model cannot predict how obstacles evolve as a function of steering actions. This posterior-prior gap persists across three input modalities (64×64 RGB, 128×128 RGB, RGBD), multiple preference formulations, auxiliary training signals, and two data strategies. We isolate the blind spot to the RSSM *transition model* specifically: the representation contains obstacle information, but the dynamics model fails to propagate it through action-conditioned rollouts. Our results demonstrate that posterior epistemic competence does not guarantee planning-level epistemic competence—a critical limitation for safety-relevant active inference.

1. Introduction

Active inference (Friston, 2010; Parr et al., 2022) provides a unified framework for perception, learning, and action through free energy minimization. A key promise is that

goal-directed behavior emerges from the agent’s generative model without hand-crafted reward functions: the agent imagines future trajectories, scores them by Expected Free Energy (EFE), and selects actions that minimize surprise relative to its preferences.

For this promise to hold in safety-critical domains like autonomous driving, the agent’s world model must be *epistemically competent*—it must know what it knows and, crucially, represent what it doesn’t know. When an obstacle appears on the road, the world model should encode this as a deviation from preferred states, and imagined trajectories should reflect whether actions resolve or worsen the situation.

We test this assumption using a deep active inference agent with a Recurrent State-Space Model (RSSM) (Hafner et al., 2023) world model, GMM preference distributions, and iCEM planning, evaluated on obstacle avoidance in CARLA 0.9.16 (Dosovitskiy et al., 2017). Our contributions:

1. We identify an **epistemic blind spot** where the RSSM posterior correctly encodes obstacle presence but the prior (imagination) cannot predict obstacle dynamics from actions.
2. We introduce a **contrastive preference model** (log-ratio of two GMMs) that achieves 37+ nats separation between obstacle-free and obstacle-present posterior latents, demonstrating that the representation *contains* obstacle information.
3. We show this posterior-prior gap is **robust across modalities**: 64×64 RGB, 128×128 RGB, and RGBD (depth camera) all exhibit the same failure pattern.
4. We demonstrate that **data quality matters but is insufficient**: collision-free obstacle demonstrations improve world model driving quality (MLD 0.15m) but do not close the imagination gap.
5. We **isolate the blind spot to the transition model** via an auxiliary obstacle prediction head that achieves

perfect posterior separation (0.000 vs 1.000) and 276 nats contrastive gap—yet avoidance remains at 0%, proving the representation is adequate but the dynamics model cannot propagate obstacle information through action-conditioned rollouts.

2. Background

2.1. Active Inference and EFE

Under active inference, an agent selects policies π that minimize Expected Free Energy:

$$G(\pi) = \sum_t \gamma^t \left[\underbrace{\mathbb{E}_q[-\log p_{\text{pref}}(z_t)]}_{\text{instrumental}} - \underbrace{H[p(o_t|s_t)]}_{\text{epistemic}} \right] \quad (1)$$

where the instrumental term drives the agent toward preferred states and the epistemic term promotes information-seeking. For obstacle avoidance, if p_{pref} assigns low probability to obstacle-present states, the instrumental term should naturally drive evasive actions.

2.2. RSSM World Model

The RSSM maintains deterministic state $h_t \in \mathbb{R}^{256}$ (GRU) and stochastic state $z_t \in \mathbb{R}^{64}$ (Gaussian):

$$\text{Posterior: } z_t \sim q(z_t|h_t, o_t) \quad (\text{filtering}) \quad (2)$$

$$\text{Prior: } \hat{z}_t \sim p(z_t|h_t) \quad (\text{imagination}) \quad (3)$$

During **filtering** (observation available), the posterior incorporates visual evidence—including obstacles. During **imagination** (planning), only the prior is available, conditioned on the deterministic state and actions. The epistemic blind spot arises when the posterior encodes information that the prior cannot reproduce.

3. Experimental Setup

Architecture. ConvEncoder (4-layer CNN, $64 \times 64 \rightarrow 256\text{D}$), RSSM (deter=256, stoch=64), ObsDecoder, StateDecoder, 5-head ensemble for epistemic uncertainty. iCEM planner (500 samples, 50 elites, 5 iterations, horizon 15).

Tasks. CARLA 0.9.16 highway: Task A (lane keeping, Town04, moderate curves) and Task B (obstacle avoidance, Town06, 2–3 static obstacles per route, 373–518m).

Training data. 96K frames obstacle-free expert driving (Town04 + Town06) plus 10–30K frames obstacle-encounter data (BehaviorAgent and scripted lane-change demonstrations).

Privileged baseline. Task B with runtime ground-truth obstacle coordinates achieves 100% avoidance (28/28 obstacles) via reflexive steer prior, motor primitives, and direc-

tional CEM penalty. All six privileged mechanisms are removed for the pure AIF experiments.

4. Results

4.1. The Preference Gap is Inverted

We fit a GMM preference model on obstacle-free expert latents and evaluate log-probability on held-out obstacle-free vs. obstacle-present latents. If the GMM correctly encodes “prefer obstacle-free states,” obstacle-present latents should have lower log-probability.

Table 1. Preference gap: $\Delta = \log p(z_{\text{clean}}) - \log p(z_{\text{obs}})$. Positive = correct direction.

Input modality	Δ (nats)	Recon ratio
64×64 RGB	−24.6	$1.62 \times$
128×128 RGB	−2.3	$1.91 \times$
64×64 RGBD	−68.1	$1.35 \times$

The gap is **inverted in all cases**: obstacle-present latents receive *higher* GMM probability. The RSSM latent space encodes driving dynamics (speed, steering, lane position) rather than scene content. Higher resolution (128×128) narrows the gap but does not reverse it. Adding depth worsens it.

4.2. Contrastive Preference Achieves Posterior Separation

We replace the single GMM with a contrastive model: $\log p_{\text{pref}}(z) = \log p_{\text{clean}}(z) - \alpha \cdot \log p_{\text{avoid}}(z)$, where p_{clean} is fitted on obstacle-free and p_{avoid} on obstacle-present latents.

Table 2. Contrastive preference: posterior separation is correct, but obstacle avoidance remains at 0%.

Scale α	Raw gap	MLD (m)	Avoidance
1.0	+37.0 nats	4.0	0% (off-road)
0.3	+9.2 nats	0.32	0%
0.2	+1.1 nats	0.17	0%
0.1	+1.0 nats	0.31	0%

At $\alpha=1.0$, the 37 nats gap is strong but overwhelms driving (MLD 4.0m). At $\alpha \leq 0.3$, driving is preserved but avoidance remains 0%. The contrastive signal correctly separates *posterior* latents but cannot influence *imagined* trajectories.

4.3. The Epistemic Blind Spot

The root cause: during CEM planning, the RSSM imagines 500 trajectories via `img_step` (prior only). These imagined latent sequences do not encode obstacle proximity because the prior cannot predict obsta-

cle appearance/disappearance from steering actions. All trajectories—straight, left, right—produce similar latent sequences in imagination.

Evidence: The 128×128 RGB model achieves 100% success rate (goal reached) with 0.17m mean lateral deviation and zero offroad events—the best driving quality of any variant—yet avoids zero obstacles. The agent drives perfectly but is blind to obstacles in its plans.

4.4. Clean Data Does Not Close the Gap

We collected 30K collision-free obstacle avoidance demonstrations (40% clean rate via waypoint-redirect autopilot) and retrained. With 10K obstacle frames (9.4% of training data), driving quality is preserved (MLD 0.15m) but avoidance remains 0%. The RSSM learns to *reconstruct* obstacle images ($1.62 \times$ higher reconstruction error) but not to *imagine* obstacle dynamics.

4.5. Auxiliary Head Isolates the Transition Model

To determine whether the blind spot lies in the *representation* (latent space lacks obstacle information) or the *transition model* (dynamics cannot propagate it), we add an auxiliary obstacle prediction head—a small MLP ($320 \rightarrow 128 \rightarrow 1$) trained with binary cross-entropy alongside the VFE loss. The head receives RSSM features and predicts obstacle presence ($\beta_{\text{aux}}=1.0$). Crucially, this head is used **only during training** to shape the latent space; at evaluation time, only the shaped representation and standard EFE scoring are used.

Table 3. Auxiliary head forces obstacle information into the latent space, achieving perfect posterior separation. Avoidance remains 0%.

Model	Obs. head (clean / obs)	Contr. gap (eff. nats)
v6 (no aux)	N/A	1.1
v10 (no aux, clean data)	N/A	1.1
v11 (aux head)	0.000 / 1.000	133.1

Obs. head: sigmoid output (0=clean, 1=obstacle). Contr. gap: contrastive log-ratio at $\alpha=0.2$.

The auxiliary head achieves **perfect binary separation** (Table 3): 0.000 for obstacle-free frames, 1.000 for obstacle-present frames. The contrastive preference gap increases from 1.1 to **133.1 effective nats** (276 raw)—a $120 \times$ improvement. This proves the latent space *can* encode obstacle presence when forced by supervision.

Yet obstacle avoidance remains at **0%**. The agent with the auxiliary-head-shaped latent space still collides with every obstacle. This result *definitively isolates the blind spot*:

- **Representation:** adequate. The latent space encodes obstacle presence with perfect fidelity.
- **Transition model:** deficient. The GRU-based dynamics $f(h_t, z_t, a_t) \rightarrow h_{t+1}$ does not learn that “steer left when obstacle ahead” produces different future latent states than “continue straight.”

During imagination, the RSSM transition model rolls out future states using only the prior. Even though the current posterior state encodes “obstacle present,” the transition model maps all action sequences to similar future states—it has not learned obstacle-conditioned dynamics.

5. Discussion

5.1. Posterior vs. Prior Epistemic Competence

Our results reveal a dissociation between two levels of epistemic competence in world-model-based active inference:

Posterior competence (filtering): The RSSM posterior correctly encodes obstacle presence when conditioned on observations. The contrastive preference achieves 37+ nats separation without auxiliary supervision, and 276 nats with an auxiliary prediction head. The representation is not the bottleneck.

Prior incompetence (imagination): The RSSM prior cannot predict how obstacle-related visual features evolve as a function of steering actions. Even when the auxiliary head forces perfect obstacle encoding into the latent space (0.000 vs 1.000), the transition model does not propagate this information through action-conditioned rollouts.

This posterior-prior gap is an **epistemic blind spot** localized in the **transition model**: the agent can recognize danger when it sees it but cannot predict danger from its plans.

In active inference terms, the generative model’s *likelihood* (observation model) encodes obstacles, but the *transition model* $f(h_t, z_t, a_t) \rightarrow h_{t+1}$ does not learn obstacle-conditioned dynamics. The auxiliary head experiment serves as a controlled ablation that separates representation adequacy from transition adequacy.

5.2. Implications for AIF Safety

This finding has direct implications for deploying active inference agents in safety-critical settings:

1. **EFE planning is only as good as imagination.** A preference model that perfectly separates safe from unsafe states is useless if the world model cannot imagine the consequences of actions in those states.
2. **Observation-level epistemic uncertainty is insufficient.** Ensemble disagreement and reconstruction er-

ror detect obstacles at the observation level, but these signals are not available during imagination (no observations).

3. **The latent bottleneck prioritizes dynamics over content.** With 64 stochastic dimensions, the RSSM encodes temporally smooth driving dynamics rather than spatially localized scene content (obstacles). This is efficient for lane keeping but blind to safety-critical objects.

5.3. Toward Closing the Gap

Three directions may address this epistemic blind spot while remaining within the active inference framework:

Larger latent space. Increasing stochastic dimensions from 64 to 256 may provide enough capacity to encode both driving dynamics and scene content. This is the most self-supervised approach but may require proportionally more training data.

Auxiliary prediction signal. We tested a supervised obstacle detection head during training that forces the latent space to encode obstacle presence (Section 4.5). While this achieves perfect posterior separation, the transition model remains obstacle-blind. A stronger version—training the auxiliary head on *imagined* rollouts rather than posterior states—may force the transition model to propagate obstacle information, but this requires differentiating through the imagination loop.

Object-centric architectures. Models like SLATE (Singh et al., 2022) or SAVi (Kipf et al., 2022) that decompose scenes into object slots may naturally separate obstacles from background, making obstacle dynamics explicitly representable in imagination.

6. Related Work

Deep active inference. Catal et al. (2020) and Fountas et al. (2020) apply active inference with VAE world models to visual navigation and Atari. Millidge (2020) connects active inference with policy gradients. Lanillos et al. (2021) surveys active inference in robotics. Our work extends to continuous-action driving and identifies a specific failure mode in RSSM-based imagination.

World models for driving. DreamerV3 (Hafner et al., 2023) uses RSSM imagination for RL-based policy learning. MILE (Hu et al., 2022) learns driving world models for imitation. Our finding—that RSSM imagination is obstacle-blind—may apply to these systems as well.

Epistemic uncertainty in planning. Chua et al. (2018) use ensemble disagreement for model-based RL. Sekar et al. (2020) explore planning to explore. Our work shows that

epistemic signals available during filtering (posterior) may not transfer to planning (prior).

7. Conclusion

We identified a fundamental epistemic blind spot in RSSM-based active inference and isolated it to the *transition model*. The world model’s posterior correctly encodes obstacle presence—a contrastive preference achieves 37 nats separation, and an auxiliary prediction head achieves perfect binary classification (276 nats)—but the prior (imagination) cannot predict obstacle dynamics from actions. This posterior-prior gap persists across input modalities (RGB, high-resolution RGB, RGBD), preference formulations (single GMM, contrastive), auxiliary supervision, and training data quality. The result is an agent that drives excellently (100% SR, 0.17m MLD at 128×128) but is completely blind to obstacles in its planning horizon.

The auxiliary head experiment provides the critical ablation: forcing obstacle information into the latent space with perfect fidelity does not help, because the GRU-based transition model $f(h_t, z_t, a_t) \rightarrow h_{t+1}$ does not learn obstacle-conditioned dynamics. This isolates the problem from representation capacity to transition model expressiveness.

Our findings highlight that posterior epistemic competence does not guarantee planning-level epistemic competence—a distinction that is critical for deploying active inference agents in safety-relevant domains. Future work should focus on transition model architectures that can learn object-conditioned dynamics, such as attention-based transitions or object-centric world models, to close this posterior-prior gap.

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